

Digital Design

Introduction to FPGA

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FPGA:-

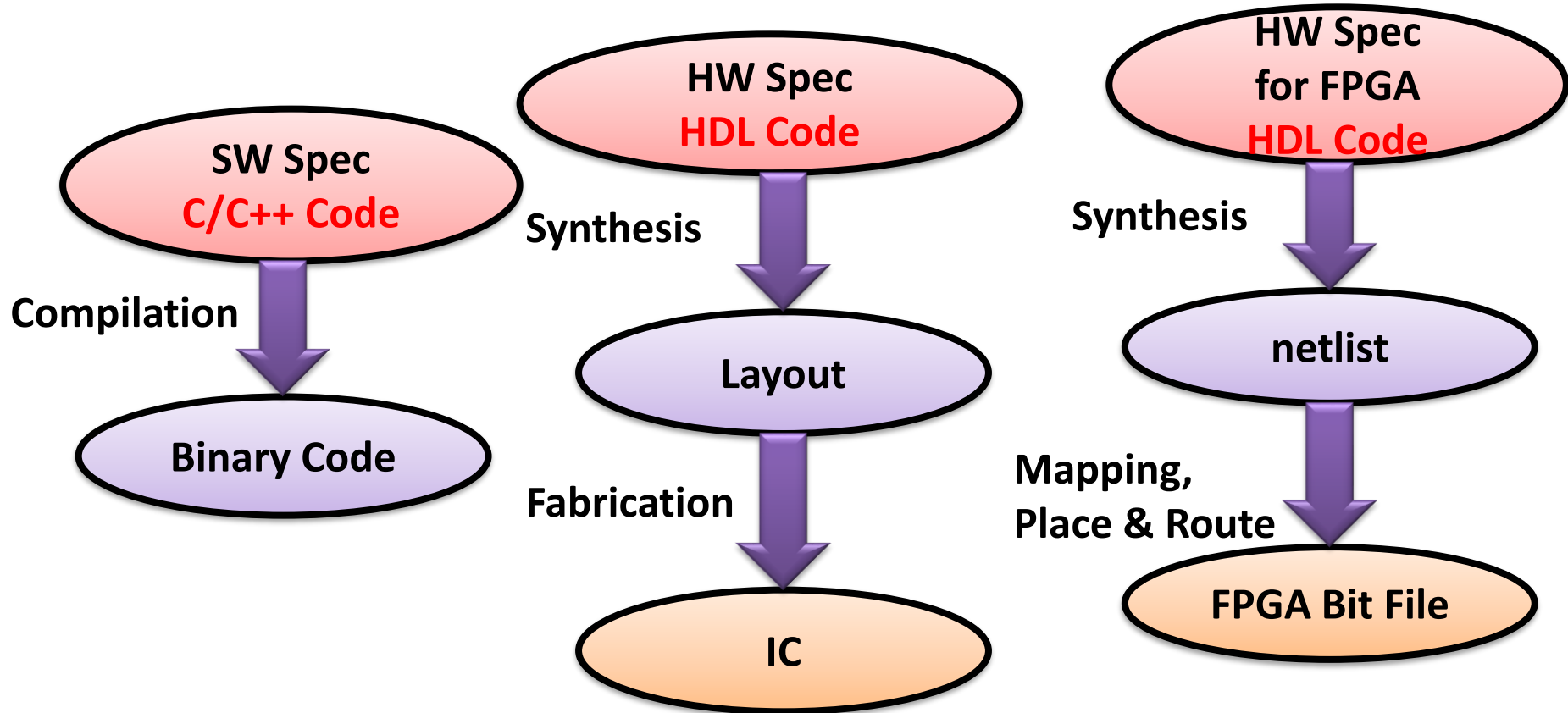
Field Programmable Gate

Array

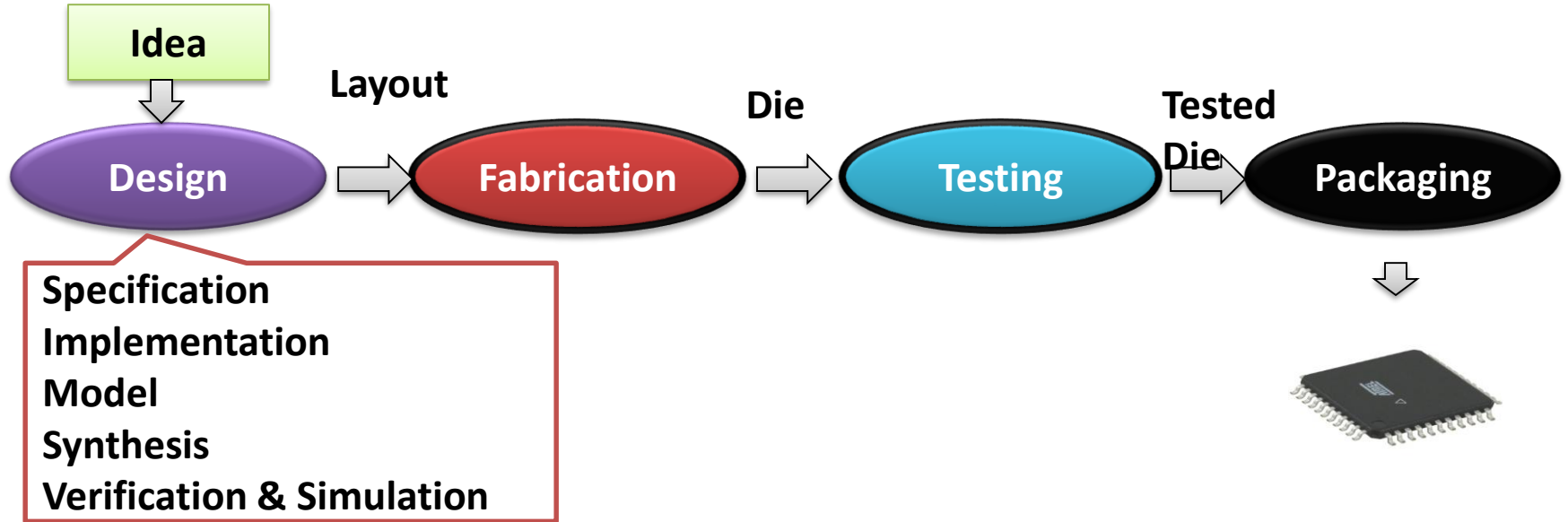
Outline

- EDA : Can we automate Digital Design Process
- IC Design : Process Flow
 - IC: ASIC
- FPGA Based Design : Process Flow
 - Configurable H/W
 - Run any digital system on top of FPGA akin to program on processor

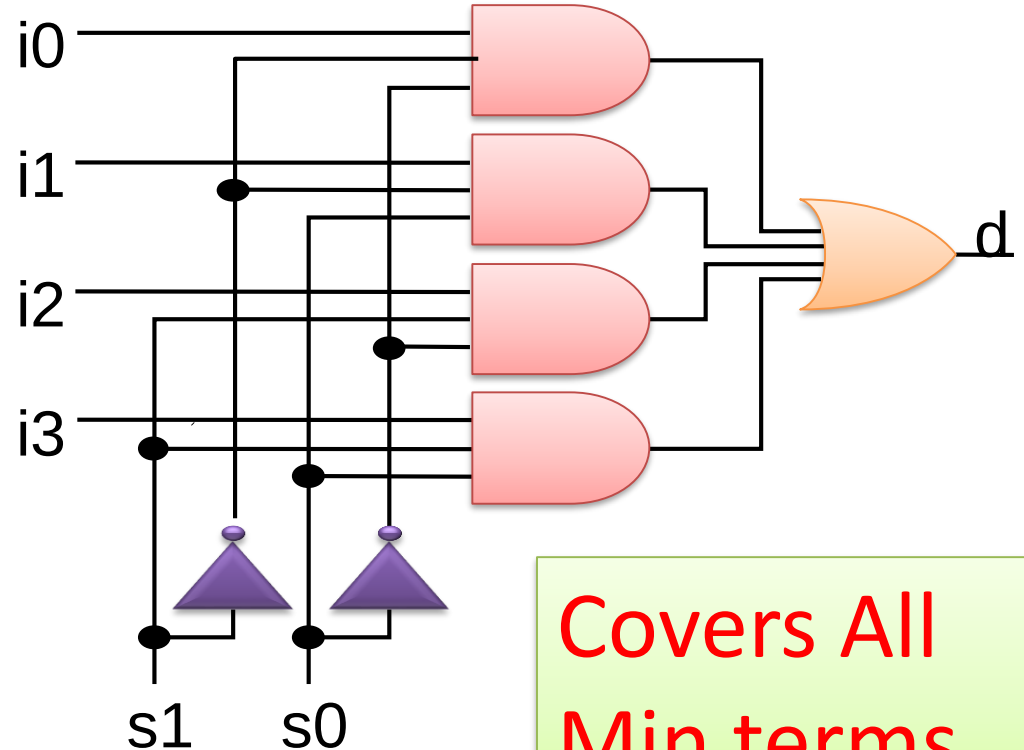
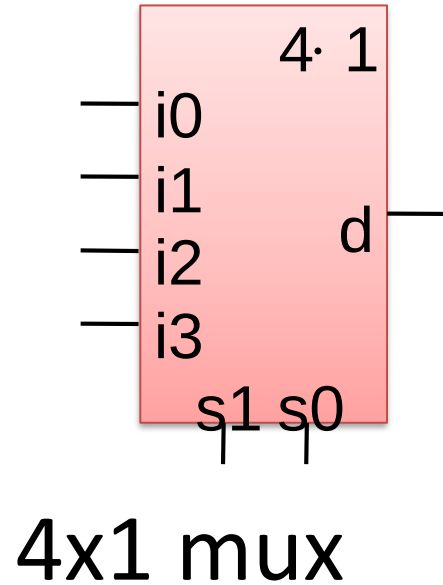
Design Flow: Hardware Vs Software Vs FPGA



IC Design Process

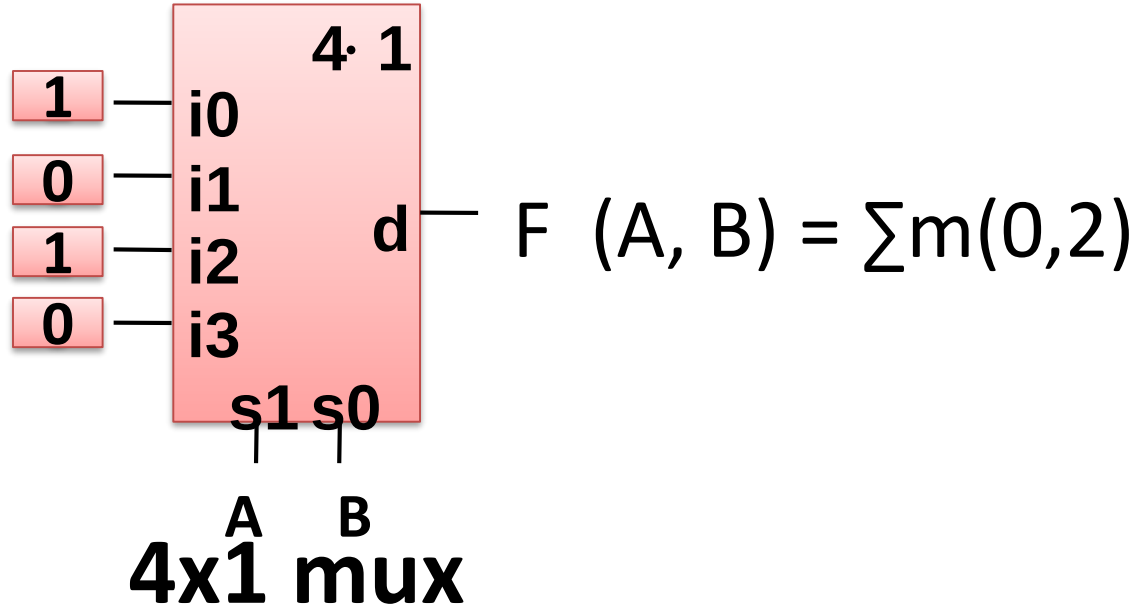


Multiplexor (Mux)



Covers All
Min terms

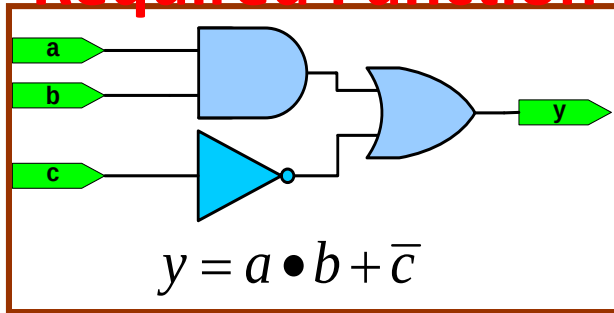
Implementing logic Function using MUX



LUT: Look up Table

- LUT is a RAM with data width of 1bit.
- The contents are programmed at power up

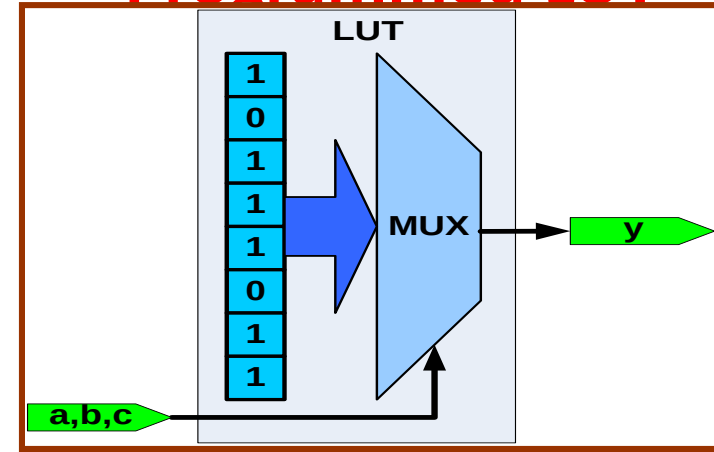
Required Function



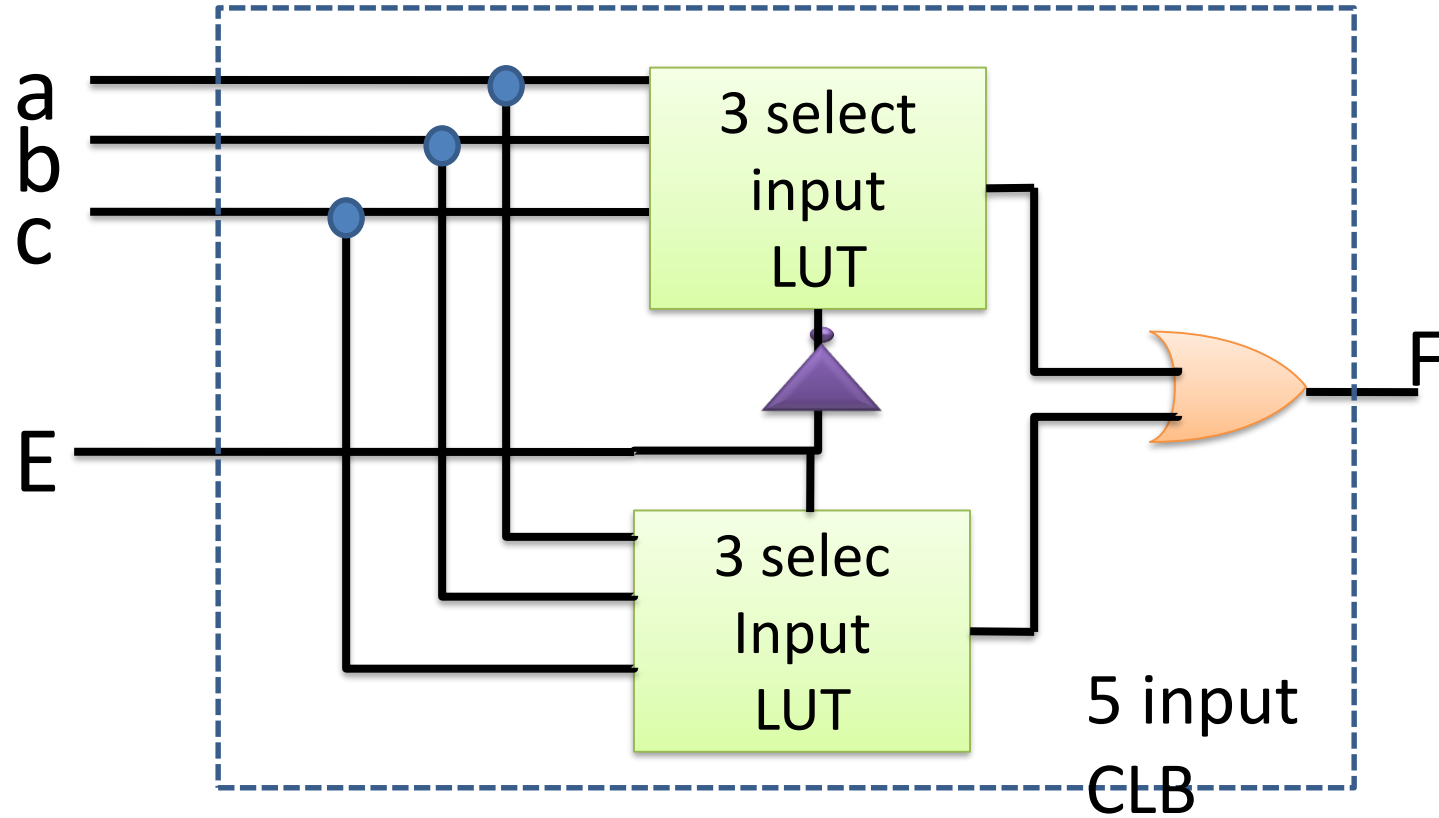
Truth Table

a	b	c	y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

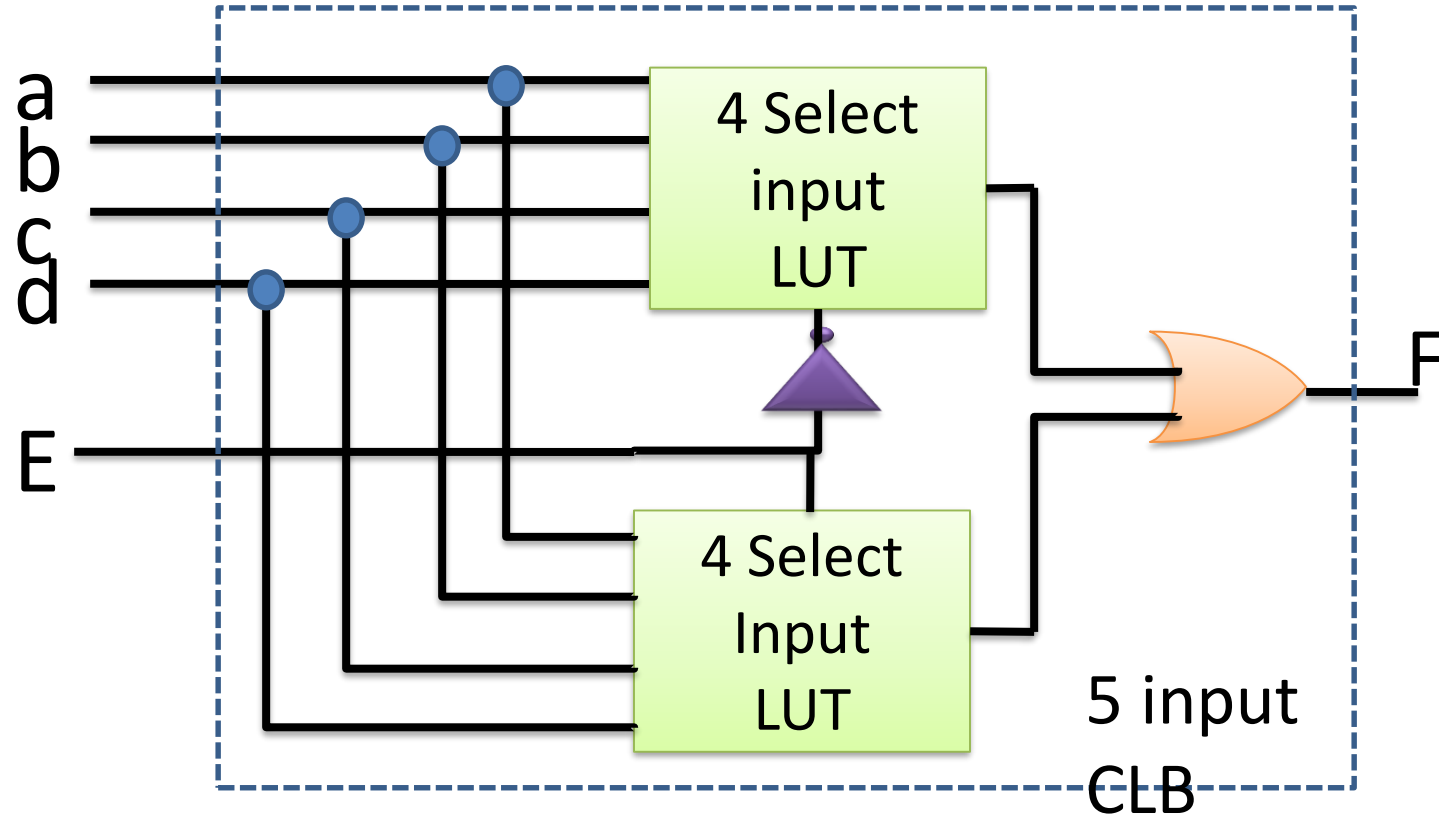
Programmed LUT



Implementing 4 Inputs Logic Function using two 3-input LUT



Implementing 5 Inputs Logic Function using two 4-input LUT



Shannon's expansion theorem

Used to implement many variable logic functions using MUX and LUTs

$$f(x_1, x_2, \dots, x_n) = x'_1 f(0, x_2, \dots, x_n) + x_1 f(1, x_2, \dots, x_n)$$

Since x_1 is a Boolean variable, we need to look at only two cases: $x_1 = 0$ and $x_1 = 1$.

- Setting $x_1 = 0$ in the above expression, we have:

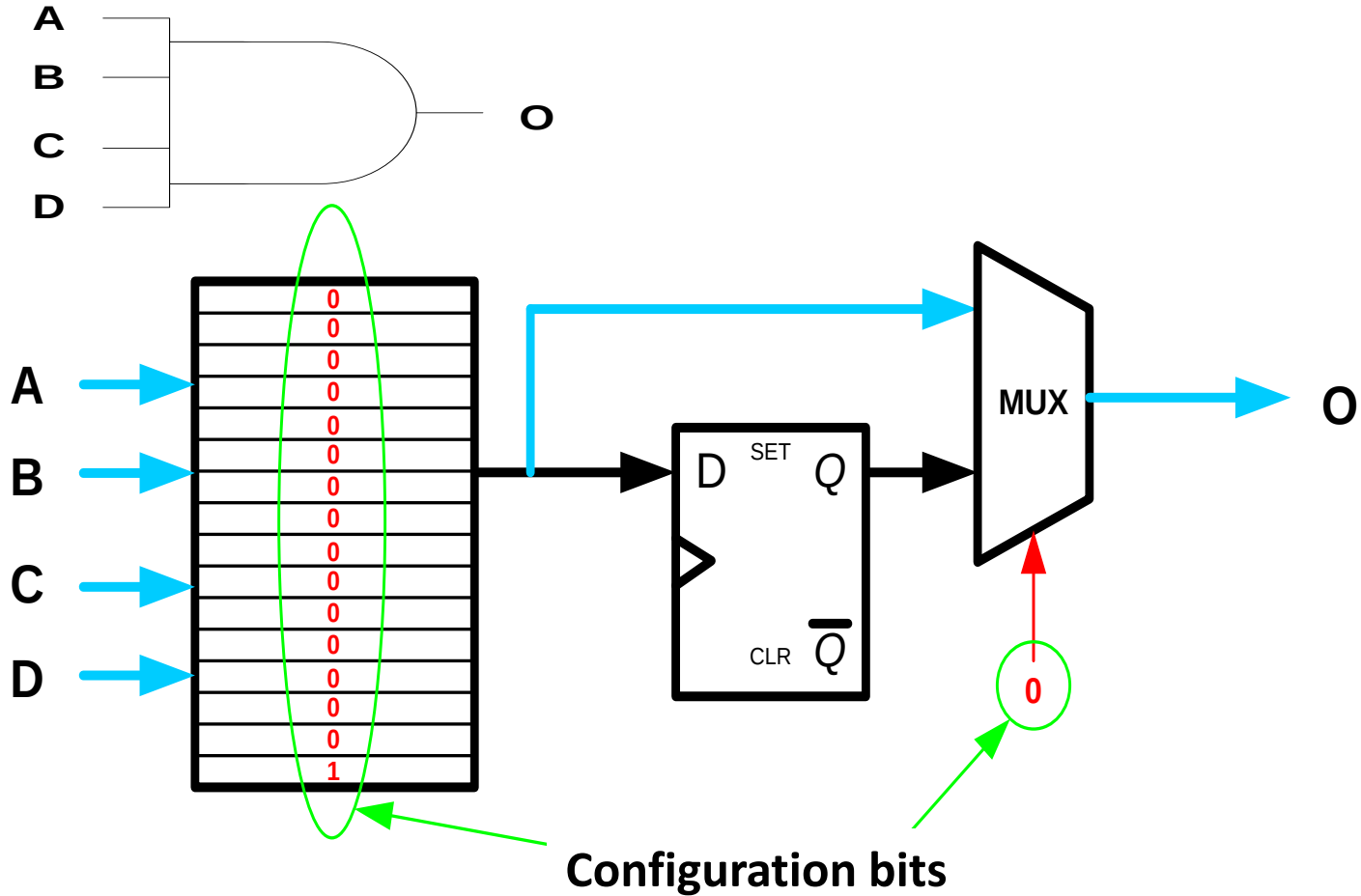
$$\begin{aligned} \bullet f(0, x_2, \dots, x_n) &= 1 f(0, x_2, \dots, x_n) + 0 f(1, x_2, \dots, x_n) \\ &= f(0, x_2, \dots, x_n) \end{aligned}$$

- Setting $x_1 = 1$, we have:

$$\begin{aligned} \bullet f(1, x_2, \dots, x_n) &= 0 f(0, x_2, \dots, x_n) + 1 f(1, x_2, \dots, x_n) \\ &= f(1, x_2, \dots, x_n) \end{aligned}$$

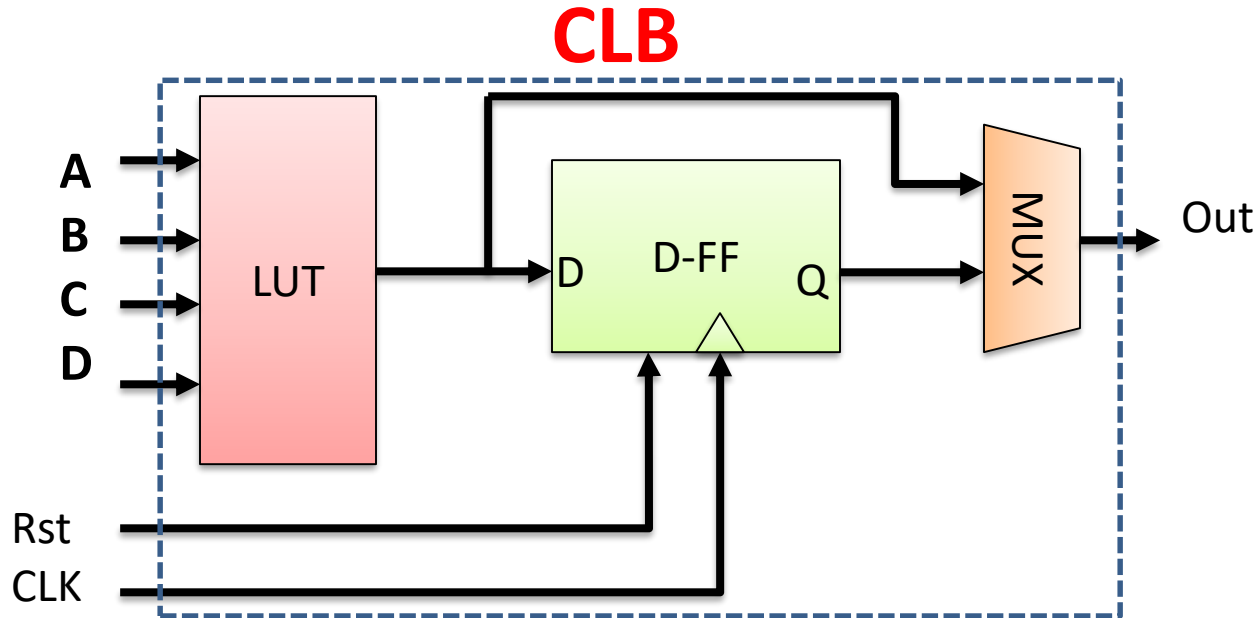
4-input AND gate with LUT

A	B	C	D	O
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1



FPGA - Field Programmable Gate Array

- Programmable logic blocks or CLB
 - Logic Element “LE”
 - Implement combinatorial & Sequential logic Based on LUT and DFF



FPGA - Field Programmable Gate Array

- Programmable logic blocks or CLB
 - Implement combinatorial and sequential logic. Based on LUT and DFF.
- Programmable I/O blocks
 - Configurable I/Os for external connections supports various voltages and tri-states.
- Programmable interconnect
 - Wires to connect inputs , outputs and logic blocks.
 - Clocks
 - short distance local connections
 - long distance connections across chip

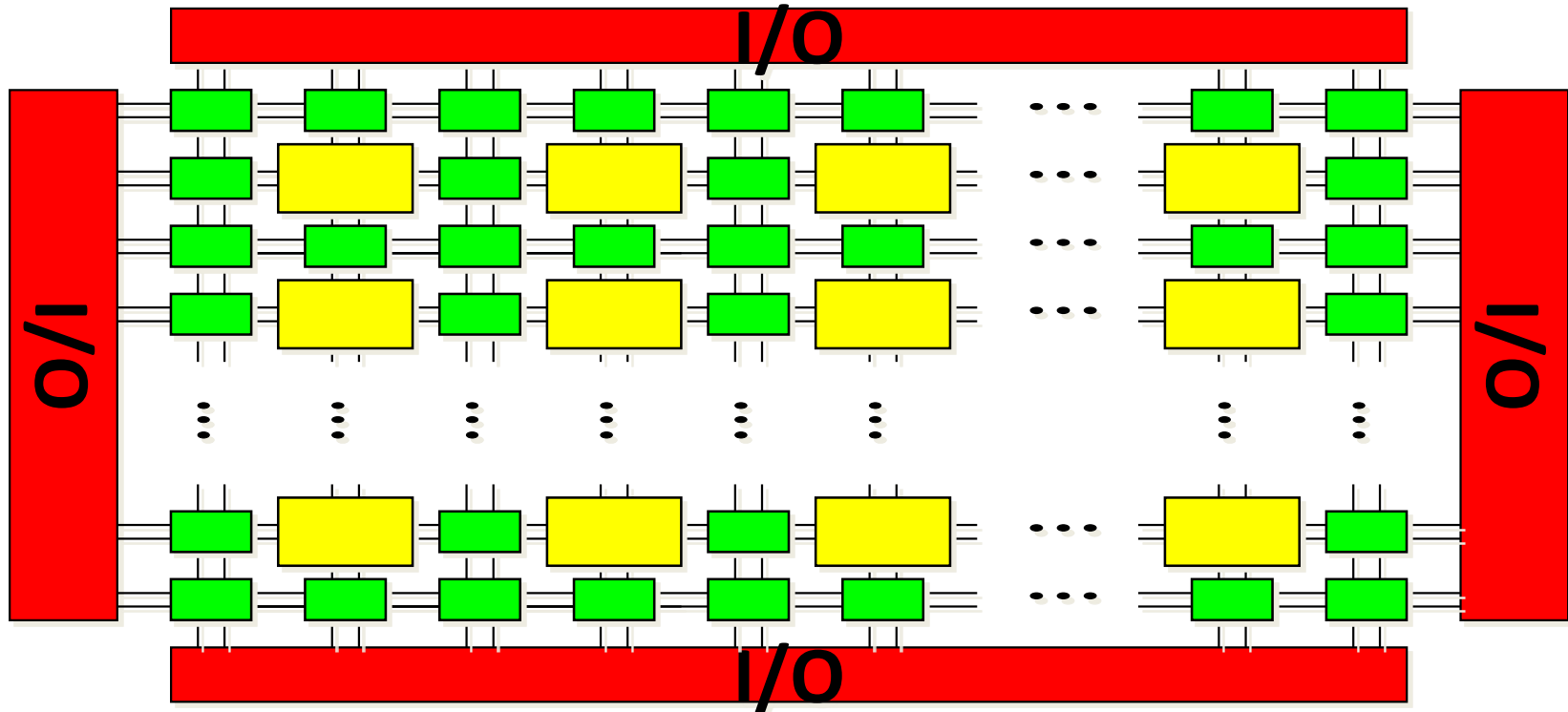
FPGA - Field Programmable Gate Array



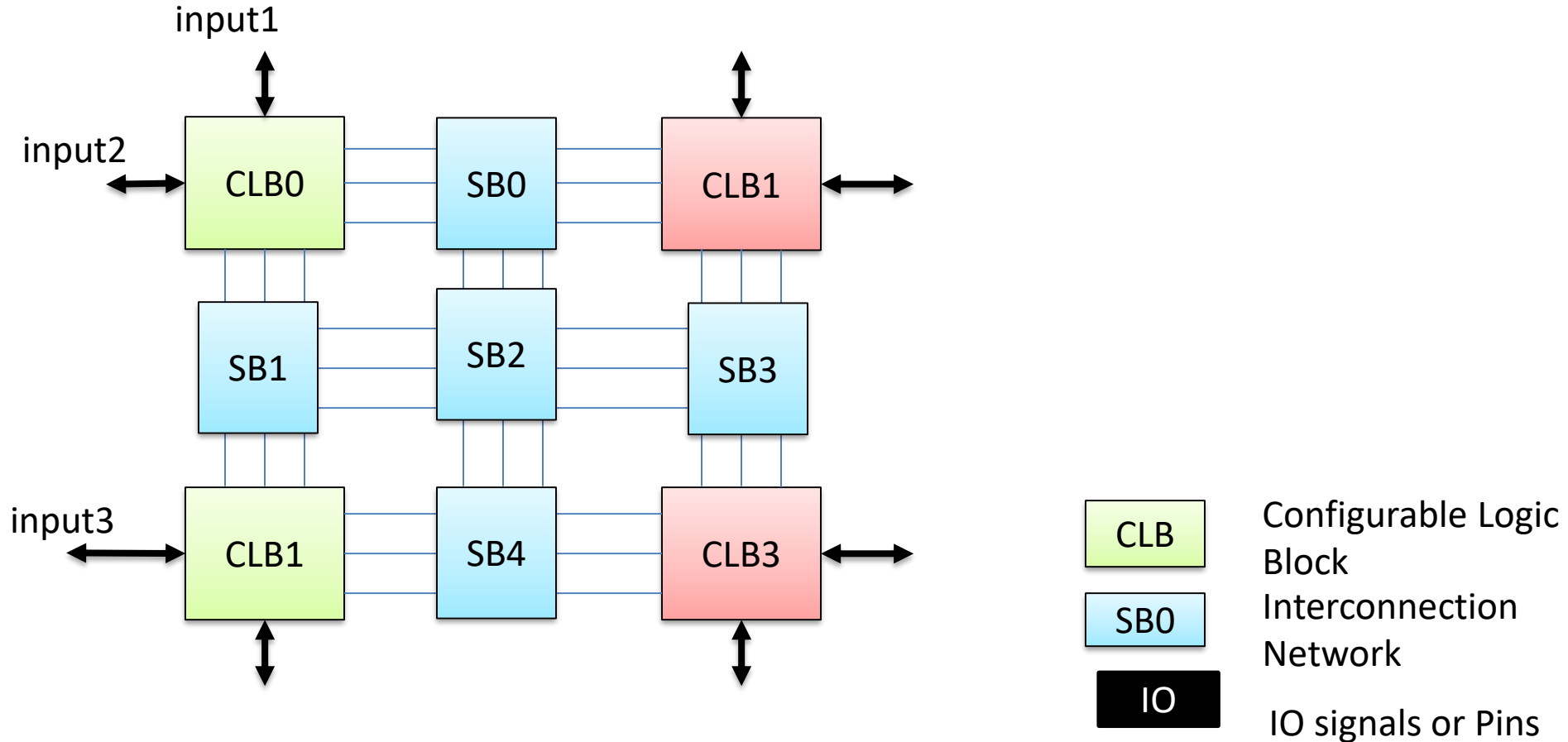
Logic block



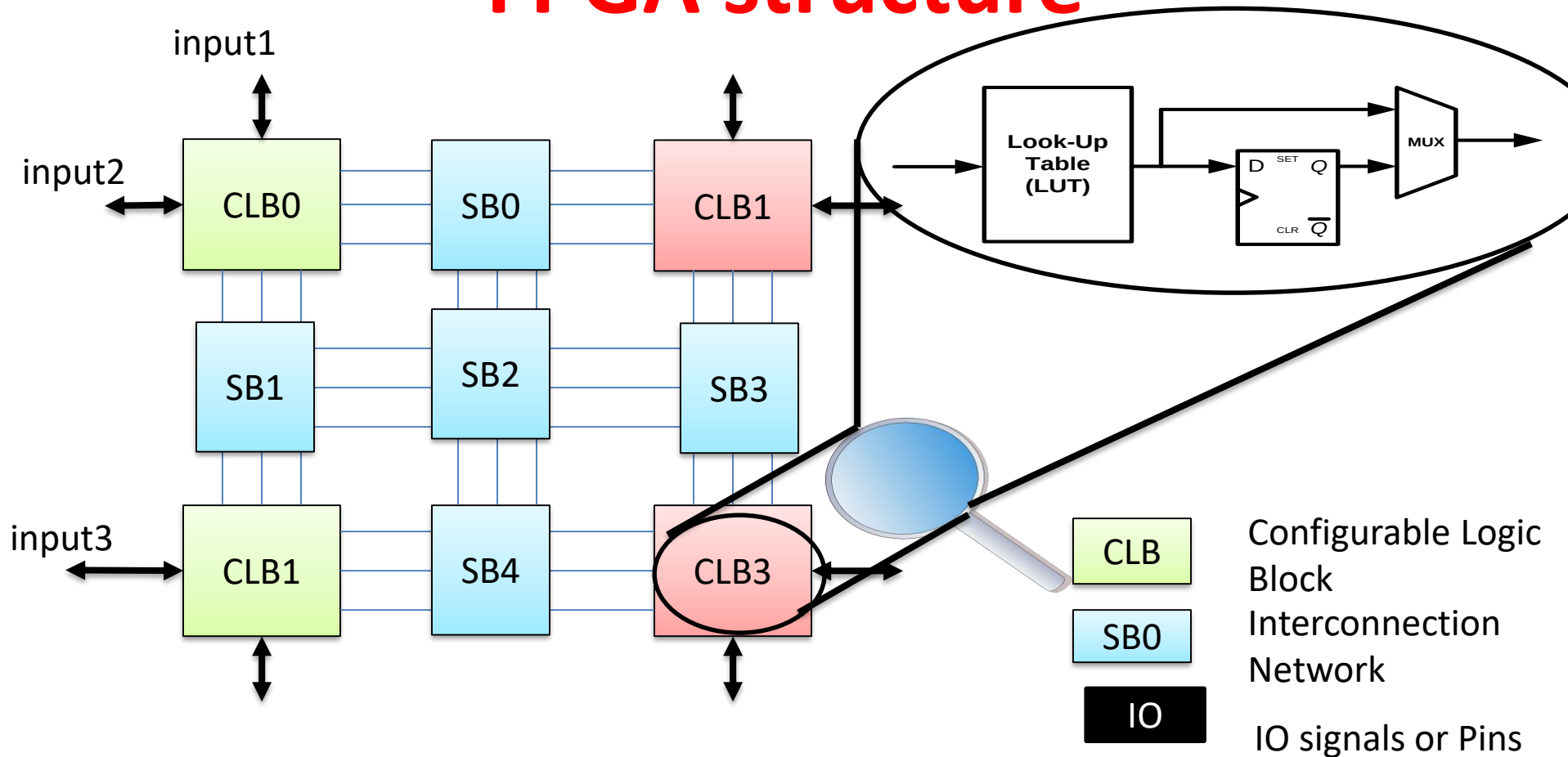
Interconnection switches



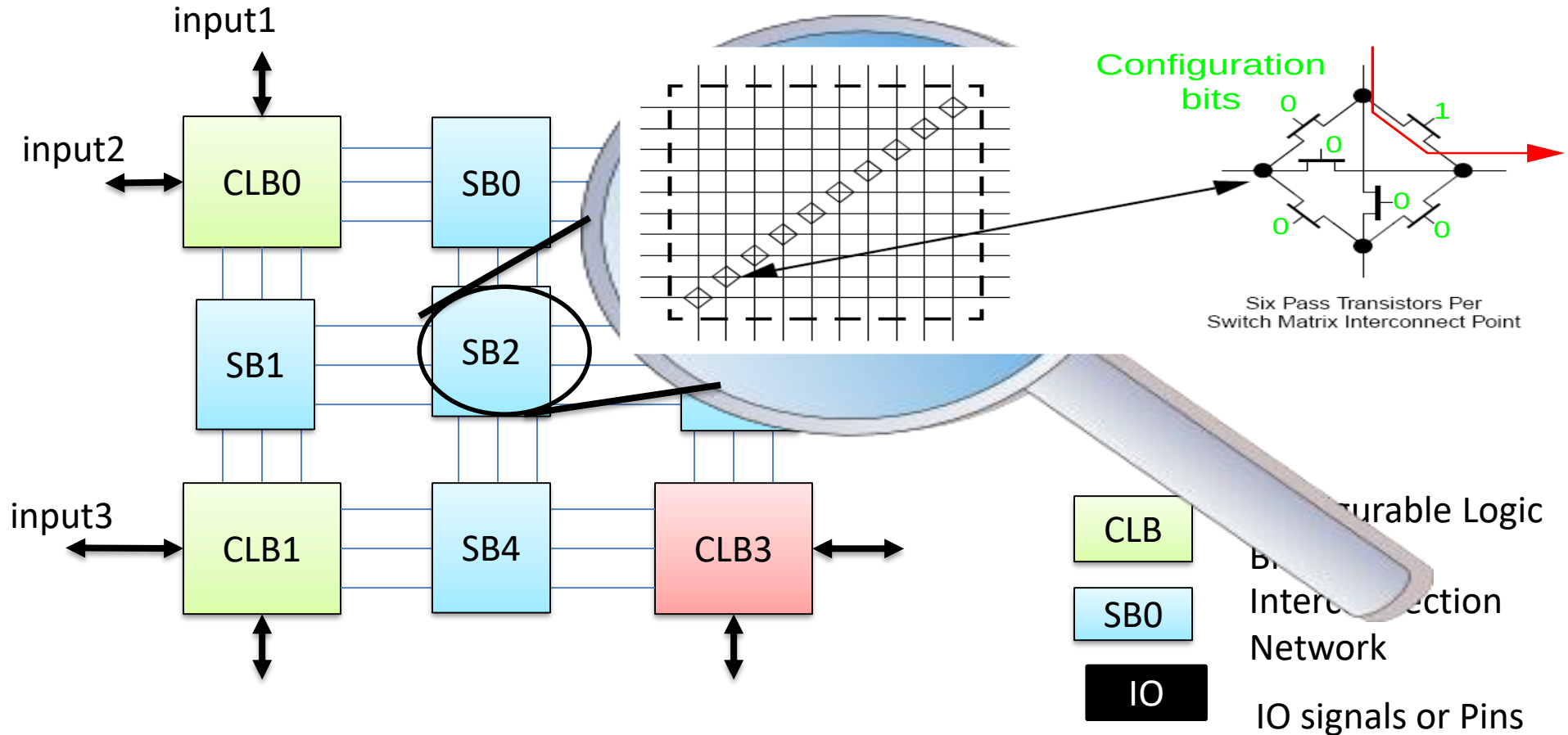
FPGA structure



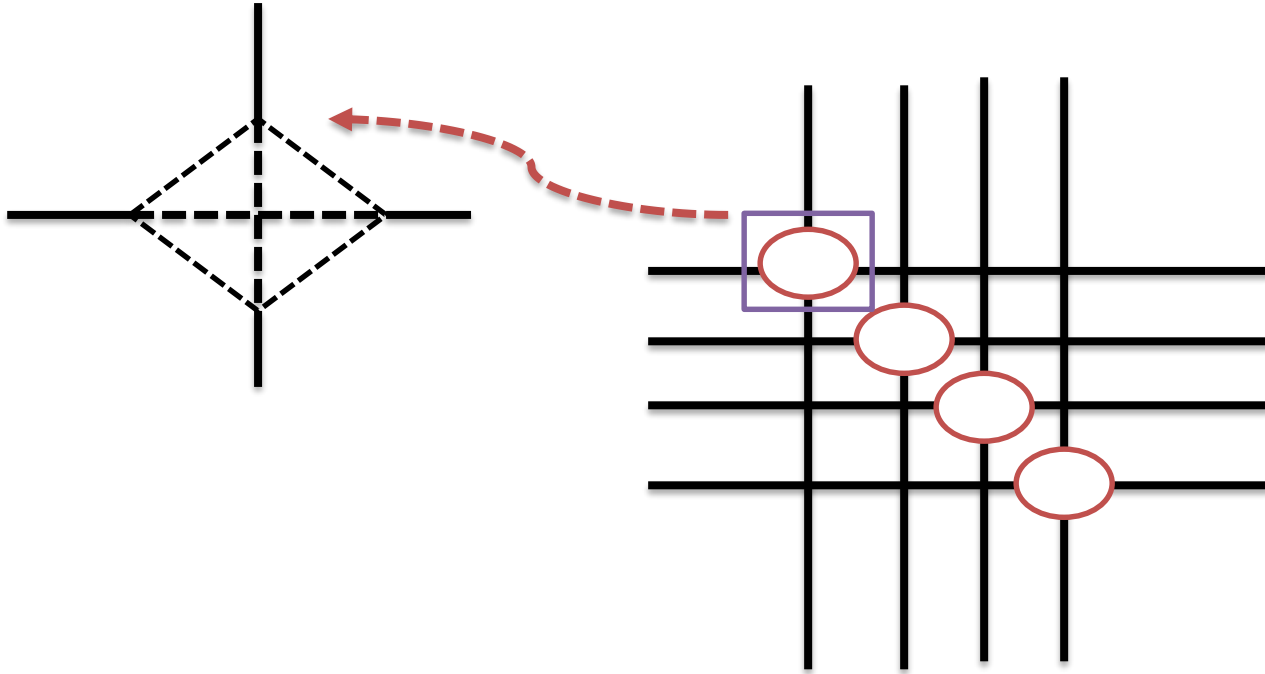
FPGA structure



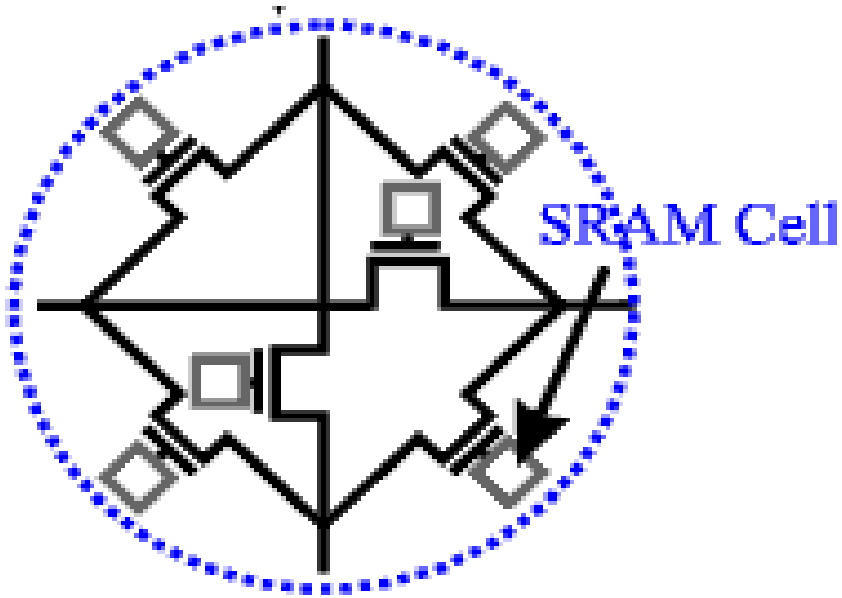
FPGA structure



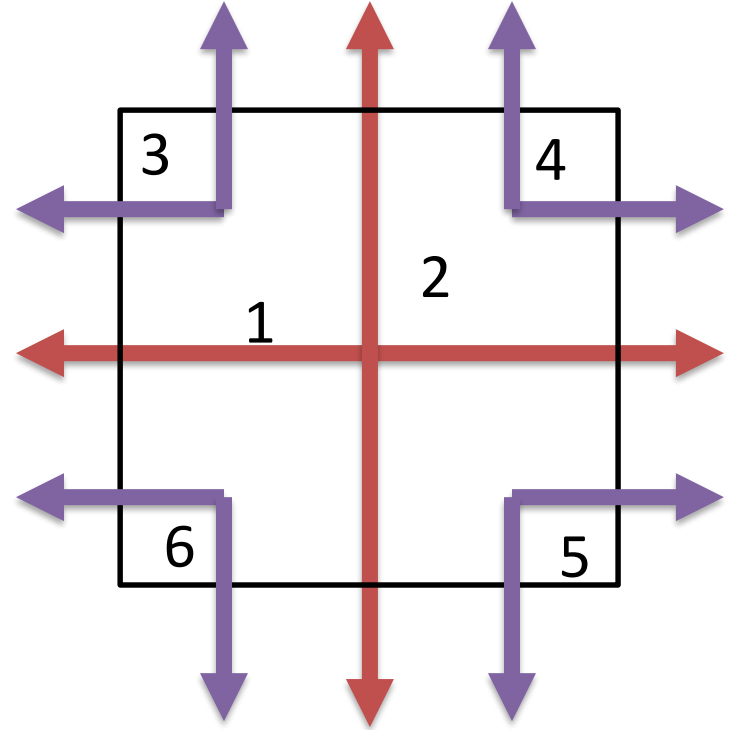
Configurable Interconnect



Configurable Interconnect

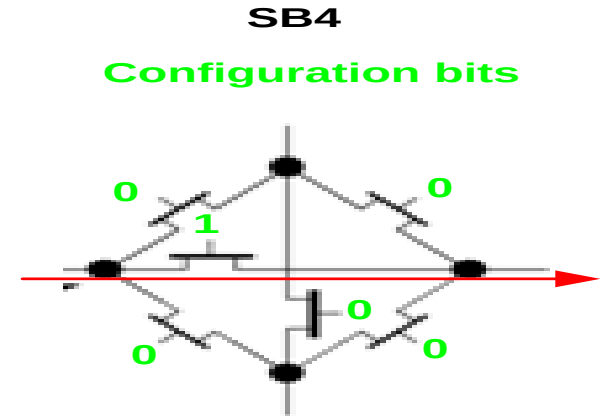
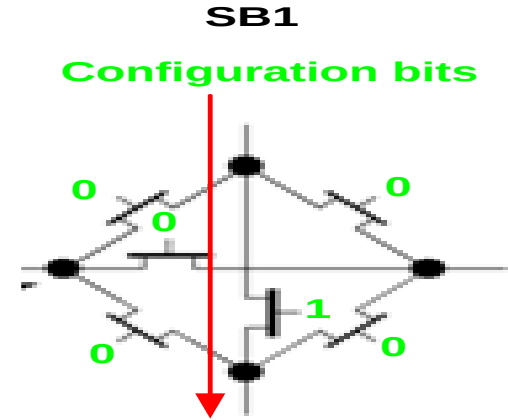
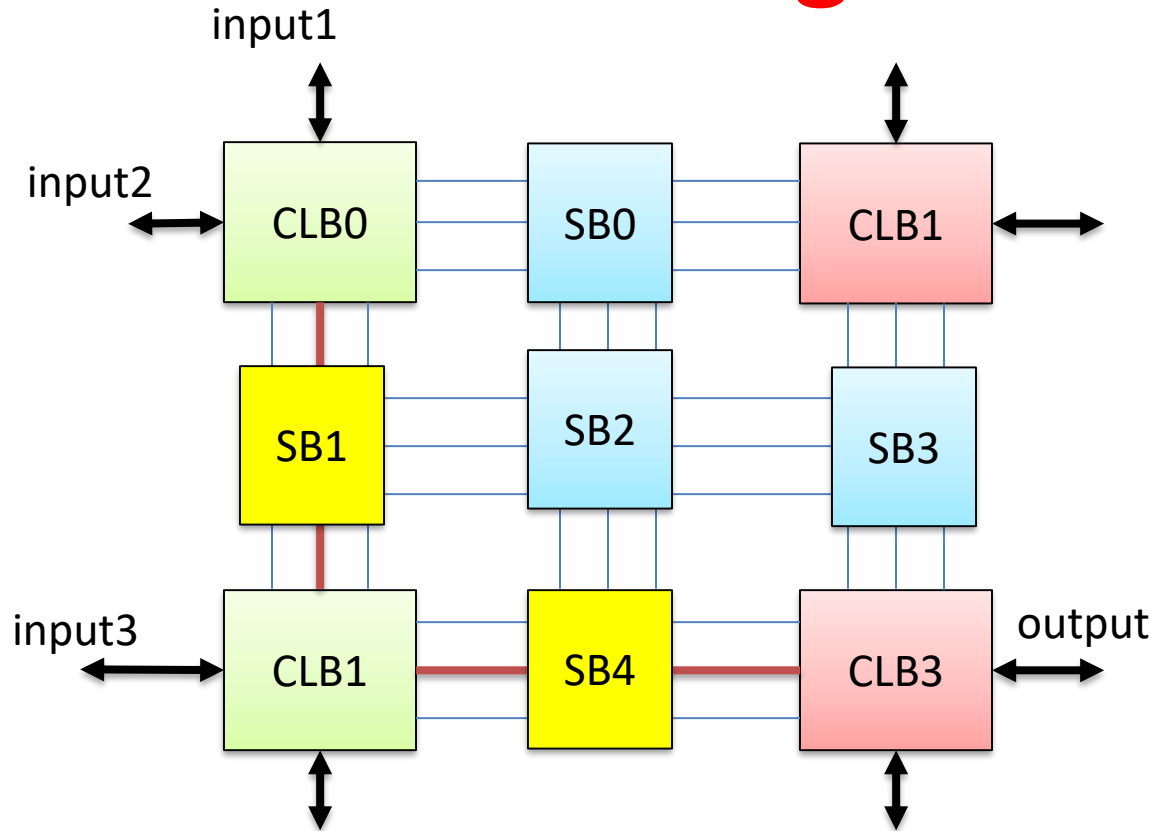


Routing Switch

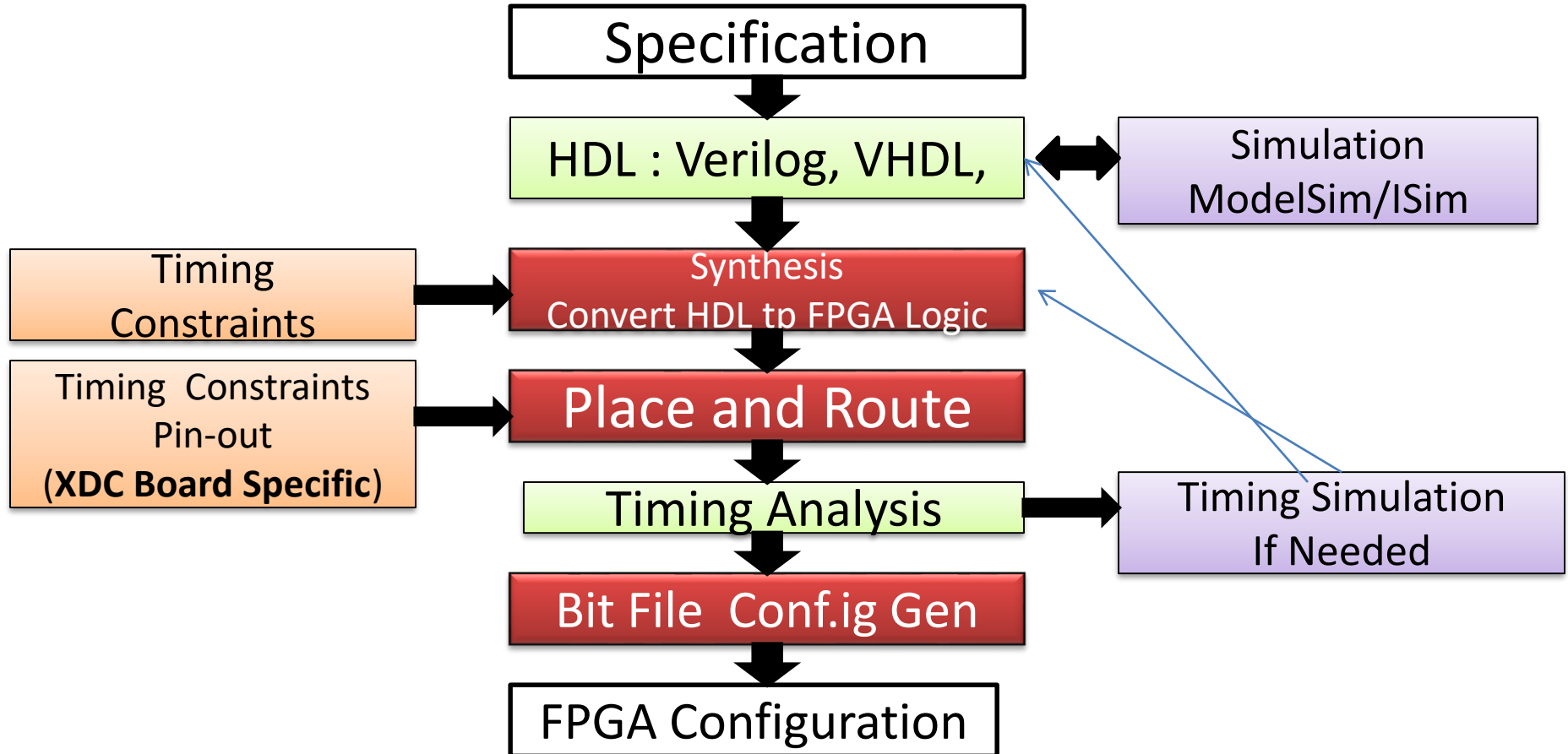


6 paths in a Switch Block: 6 bit configurations for **EW**, **NS**, **NE**, **NW**, **SE**, **SW**

Routing: Select Path



FPGA Design flow



FPGA Advantages

- **Designing with FPGA: Faster, Cheaper**
 - Quickly test our design, Easy for Educational setup (Courses)
- **Ideal for customized designs**
 - Product differentiation in a fast-changing market
- **Offer the advantages of high integration**
 - High complexity, density, reliability
 - Low cost, power consumption, small phy. size
- **Avoid the problems of ASICs**
 - high NRE cost, long delay in design and testing
 - increasingly demanding electrical issues

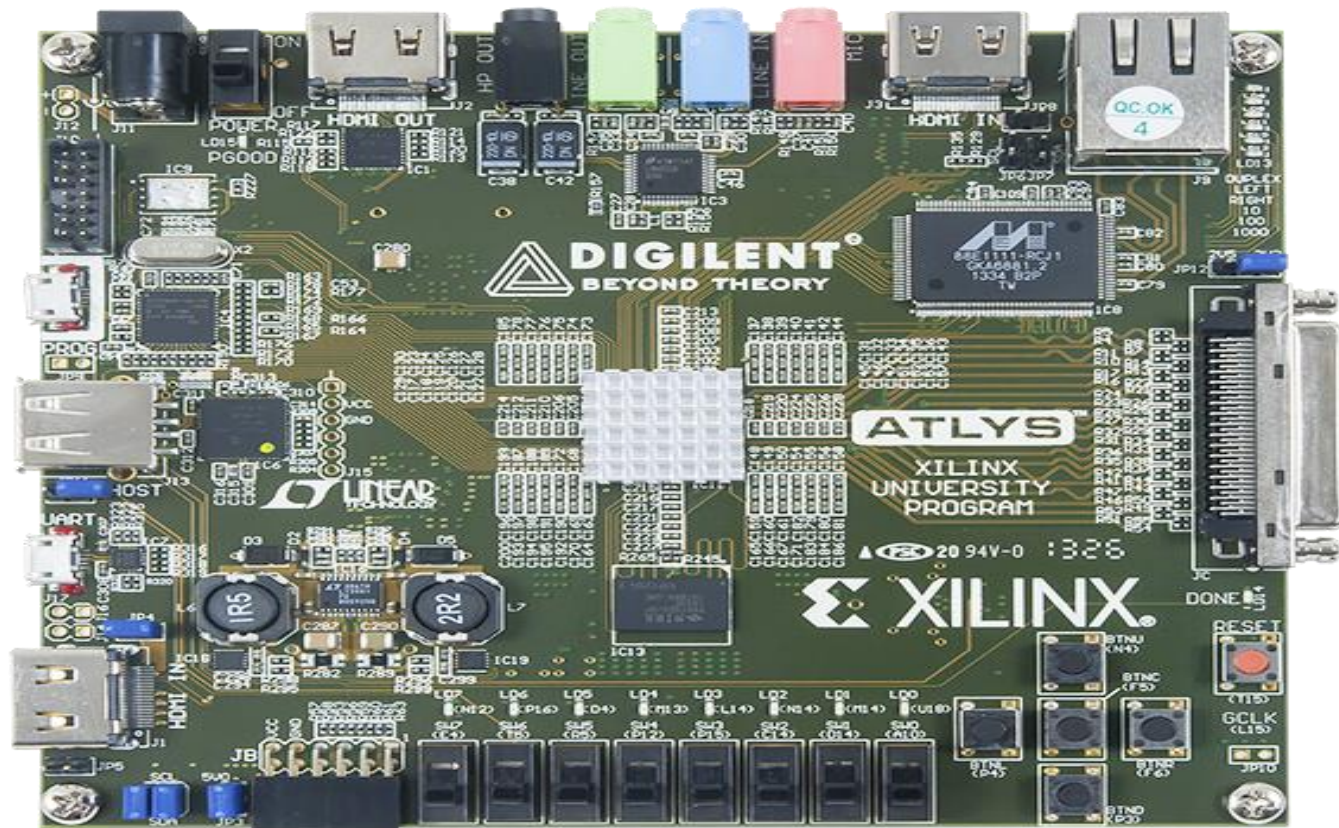
Field Programmable Device

- **Basic Section of FPD:**
 - Logical Block, Routing (Switch Matrix)
 - Input Output Block
- **More Advanced FPD Contains:**
 - On-chip Memory, Embedded Processor
 - Clock Management, High-Speed Transceiver

Digilent Xilinx Atlys FPGA Board

- [Xilinx Spartan-6](#) LX45 FPGA,
- 6,822 slices : four 6-input LUTs and eight flip-flops
- 2.1Mbits of block RAM, 128 MB DDR2
- 58 DSP slices
- JTAG programming , RJ-45 Ethernet port, 4 HDMI video ports
- AC-97 Audio Codec mic, & headphone
- USB-UART and USB-HID port
- GPIO includes 8 LEDs, 6 buttons, and 8 slide switches

Digilent Xilinx Atlys FPGA Board



Digilent Xilinx Basys 3 FPGA Board

- [Xilinx Artix-7 FPGA](#)
- 33,280 logic cells in 5200 slices (four 6-input LUTs and 8 flip-flops)
- 1,800 Kbits of fast block RAM
- 90 DSP slices
- ADC, USB-JTAG port, USB-UART Bridge , 12-bit VGA output
- USB HID Host for mice, keyboards
- 16 user switches, 16 user LEDs, 5 user pushbuttons , 4-digit 7-segment display

Using FPGA to prototype/test

- Use Xilinx ISE/Vivado
- Create new project
 - with IC number of Basys 3 FPGA, Verilog Code,
- Write Code for Design : FA
 - Synthesize and Simulate in Vivado
- Write IO constraints : XDC file
- Implement: Place and Route, Bit file generation
- Download/program the FPGA

Verilog Code for Full Adder

```
module full_adder_s ( input a,b,cin,   output sum,carry );  
    wire w1,w2,w3,w4;    //Internal connections  
    xor(w1,a,b);  
    xor(sum,w1,cin);      //Sum output  
    and(w2,a,b);  
    and(w3,b,cin);  
    and(w4,cin,a);  
    or(carry,w2,w3,w4);   //carry output  
endmodule
```

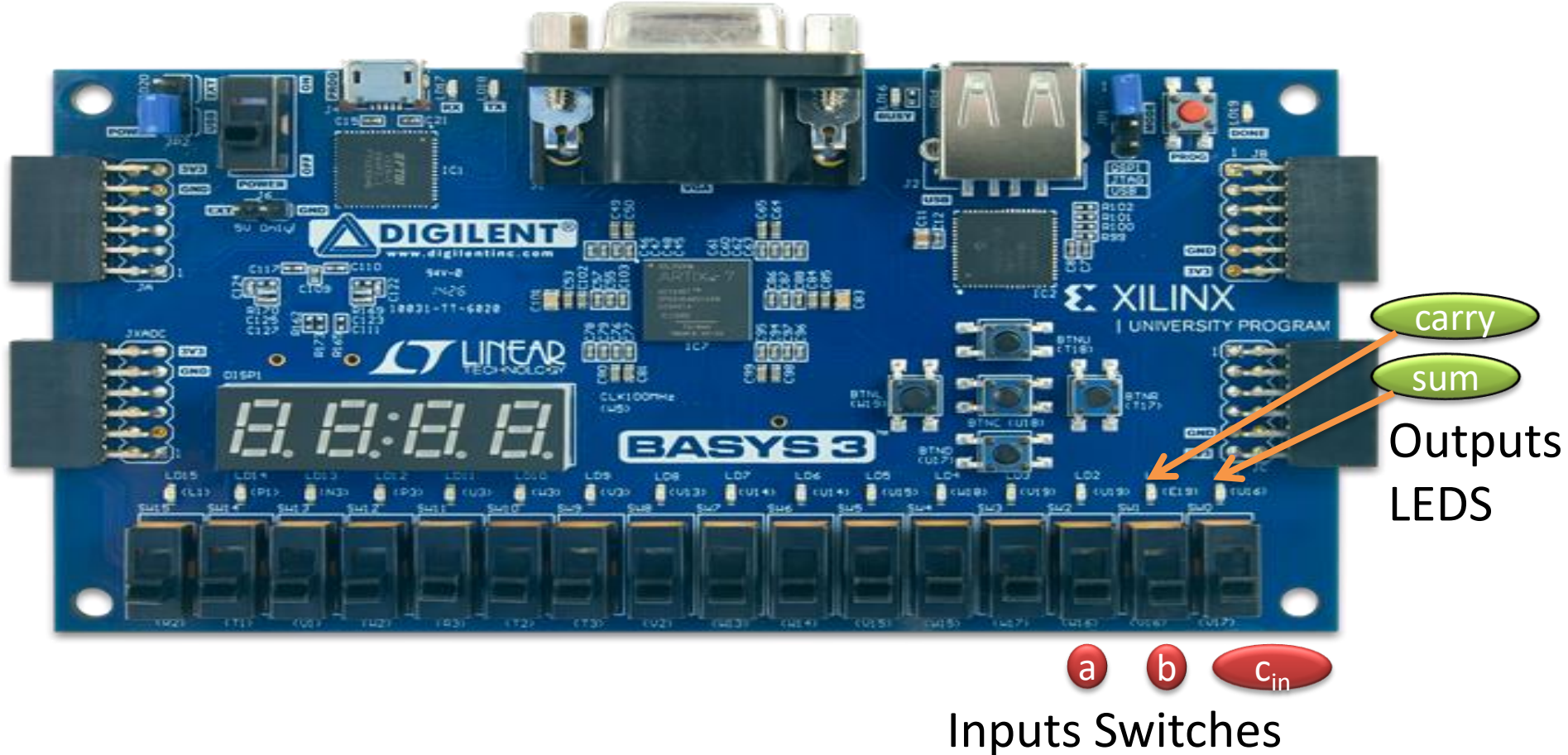
XDC File Full Adder: Basys 3

```
set_property PACKAGE_PIN V17 [get_ports {a}]
set_property PACKAGE_PIN V16 [get_ports {b}]
set_property PACKAGE_PIN W16 [get_ports {cin}]

set_property PACKAGE_PIN U16 [get_ports {sum}]
set_property PACKAGE_PIN E19 [get_ports {carry}]

set_property IOSTANDARD LVCMOS33
[get_ports {a, b, cin, sum, carry}]
```

Digilent Xilinx Basys 3 FPGA Board



Thanks